

Problem Set 1: From Two-Good Demand to Verified General Demand

ECON 803 - Microeconomic Theory I - Fall 2026

Student edition - questions only

The set has two parts. Complete Part I by hand before using a coding or formal-proof assistant. In Part II, you may use Codex, but you must inspect and validate the resulting artifacts. No due date is stated in this source; use the date announced in class.

Part I. By hand

The three board problems use numbers chosen to keep the arithmetic transparent. For each one, show the derivation and state the economic point in one sentence.

1. Nominal scaling is not a real change

Let $X = \mathbb{R}_+^2$ and $B(p, w) = \{x \in X : p \cdot x \leq w\}$.

- (a) Prove directly that $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$.
- (b) Explain why any demand correspondence generated by maximizing a fixed preference relation is homogeneous of degree zero in (p, w) . State whether continuity, concavity, or uniqueness is needed for this conclusion.
- (c) At $p = (1, 2)$ and $w = 12$, derive demand for $u(x) = \sqrt{x_1 x_2}$. Check the budget and then verify directly that multiplying prices and wealth by five leaves the chosen bundle unchanged.

2. Curvature changes substitution

Let

$$u(x) = \left(\frac{1}{2}\sqrt{x_1} + \frac{1}{2}\sqrt{x_2}\right)^2,$$

the equal-weight CES utility with $\rho = 1/2$ and $\sigma = 2$.

- (a) At $p = (1, 4)$ and $w = 10$, use the tangency condition and the budget to derive demand. Compute expenditure shares.
- (b) Raise p_2 from 4 to 9, holding $p_1 = 1$ and $w = 10$. Recompute demand and shares.
- (c) At the original prices and wealth, compute demand for the equal-weight Cobb-Douglas utility $u_{CD}(x) = \sqrt{x_1 x_2}$. Explain what the comparison isolates.

3. Corners and kinks are economic, not errors

At $p = (1, 2)$ and $w = 12$, consider separately $u^A(x) = 2x_1 + x_2$ and $u^B(x) = \min\{x_1, x_2\}$.

- (a) Derive each demand.
- (b) For each model, identify the inequality or equality that determines the solution.
- (c) Explain why applying an interior differentiable tangency condition mechanically would miss the point of both examples.

Part II. Verified scale-up with Codex and Lean

Part II extends Problems 1 and 2. It does not replace them. Complete the bridge before opening Codex.

4. The bridge

Write four short answers:

- 1. Seed: Which conclusion from Problem 1 or 2 will you extend?
- 2. Fixed: Which preference, budget, and positivity assumptions remain unchanged?
- 3. Scaled: Which object changes from two goods to five goods or arbitrary finite dimension?
- 4. Risk: Which domain violation, numerical error, or logical overclaim becomes possible after scaling?

5. Computational scale-up: five-good CES demand

For L goods, normalized weights $a_\ell > 0$ with $\sum_\ell a_\ell = 1$, and elasticity $\sigma > 0$, use

$$b_\ell(p) = \frac{a_\ell^\sigma p_\ell^{1-\sigma}}{\sum_k a_k^\sigma p_k^{1-\sigma}}, \quad x_\ell(p, w) = \frac{w b_\ell(p)}{p_\ell}.$$

Take

$$a = (0.10, 0.15, 0.20, 0.25, 0.30), \quad p = (1, 2, 3, 4, 5), \quad w = 120,$$

and compare $\sigma \in \{1/2, 1, 2\}$.

(a) Before coding, derive the displayed share formula from the CES first-order conditions and say which normalization makes it valid at $\sigma = 1$.

(b) Ask Codex to create `work/ps01_ces.py`. The program must return shares and quantities, validate dimensions and strict positivity, retain full precision internally, and print one readable comparison table.

(c) Add automated checks for $\sum_\ell b_\ell = 1$, $p \cdot x = w$, homogeneity under $(p, w) \mapsto (7p, 7w)$, and the Cobb-Douglas share restriction at $\sigma = 1$. Use an explicit numerical tolerance no larger than 10^{-10} .

(d) Add one deliberately rejected input with a zero price. Explain why rejection, rather than a numerical answer, is the economically correct behavior.

(e) Interpret how expenditure-share dispersion changes with σ . This paragraph must be written by you; a finite table is not a general proof.

A suitable first prompt is:

Read Problems 2 and 5 and problemsets/AGENTS.md. First restate the CES assumptions and propose a dependency-ordered plan. Then implement only the five-good scale-up in work/ps01_ces.py, add the specified identity checks and rejected input, run the file, and report the command and result. Do not write the final economic interpretation.

6. Formal scale-up: arbitrary finite dimension

Problem 1 proved a two-good budget identity. The formal target removes the arbitrary two-good restriction while retaining the economically essential positive common scale.

(a) Read formal/Econ803/Consumer/Demand.lean and Starter.lean. Record the types of InBudget, UtilityDemand, inBudget_scale_iff, and utilityDemand_scale_iff.

(b) In your own work/PS01_ConsumerDemand.lean, prove the proposition TwoScaleClaim: for arbitrary finite L , utility u , bundle x , price vector p , wealth w , and positive scalars a, b , the same bundle is demanded at (ap, aw) if and only if it is demanded at (bp, bw) .

(c) Do not use sorry, admit, a new axiom, or a numerical specialization. Ask Codex first for the dependency map, then for a proof using the existing one-scale theorem.

(d) From formal/, validate with

```
lake env lean ../problemsets/ps01_consumer_demand/work/PS01_ConsumerDemand.lean
```

Run #print axioms on your completed theorem and report the output separately from compilation success.

(e) Explain why this theorem needs neither differentiability nor concavity, and why it does not by itself prove budget exhaustion. Name the additional economic assumptions used by the book's budget-exhaustion theorem.

7. Submission audit

Submit the handwritten Part I, the completed four-line bridge, material Codex prompts, both source artifacts, validation output, the rejected-input test, and a short audit naming one generated step you checked independently. End with one paragraph separating:

- what you established by hand;
- what computation checked on the specified inputs;
- what Lean proved for arbitrary finite dimension; and
- what remains an economic assumption or interpretation.